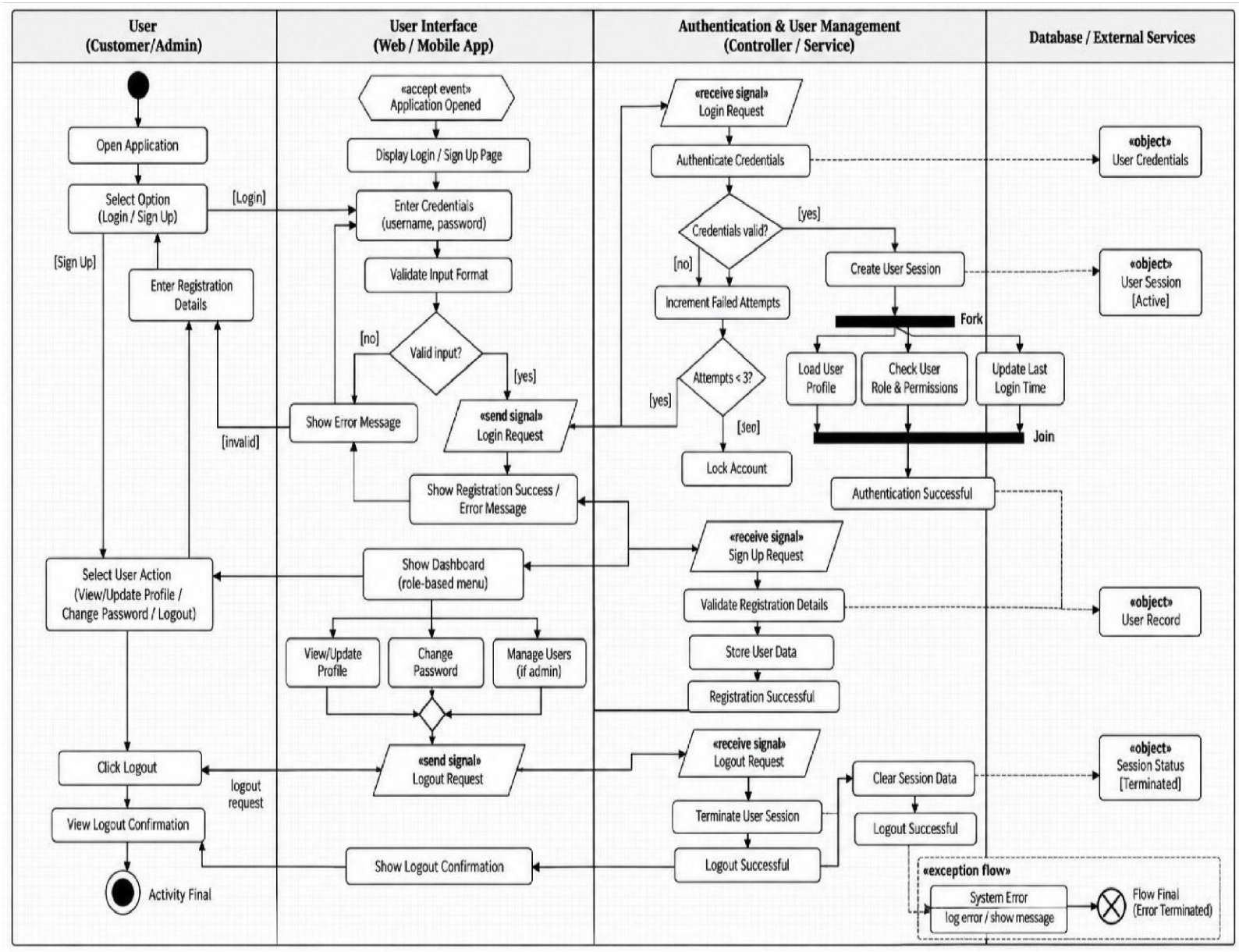
OOAD LAB ASSESSMENT - 5

Ticket Booking and Reservation System

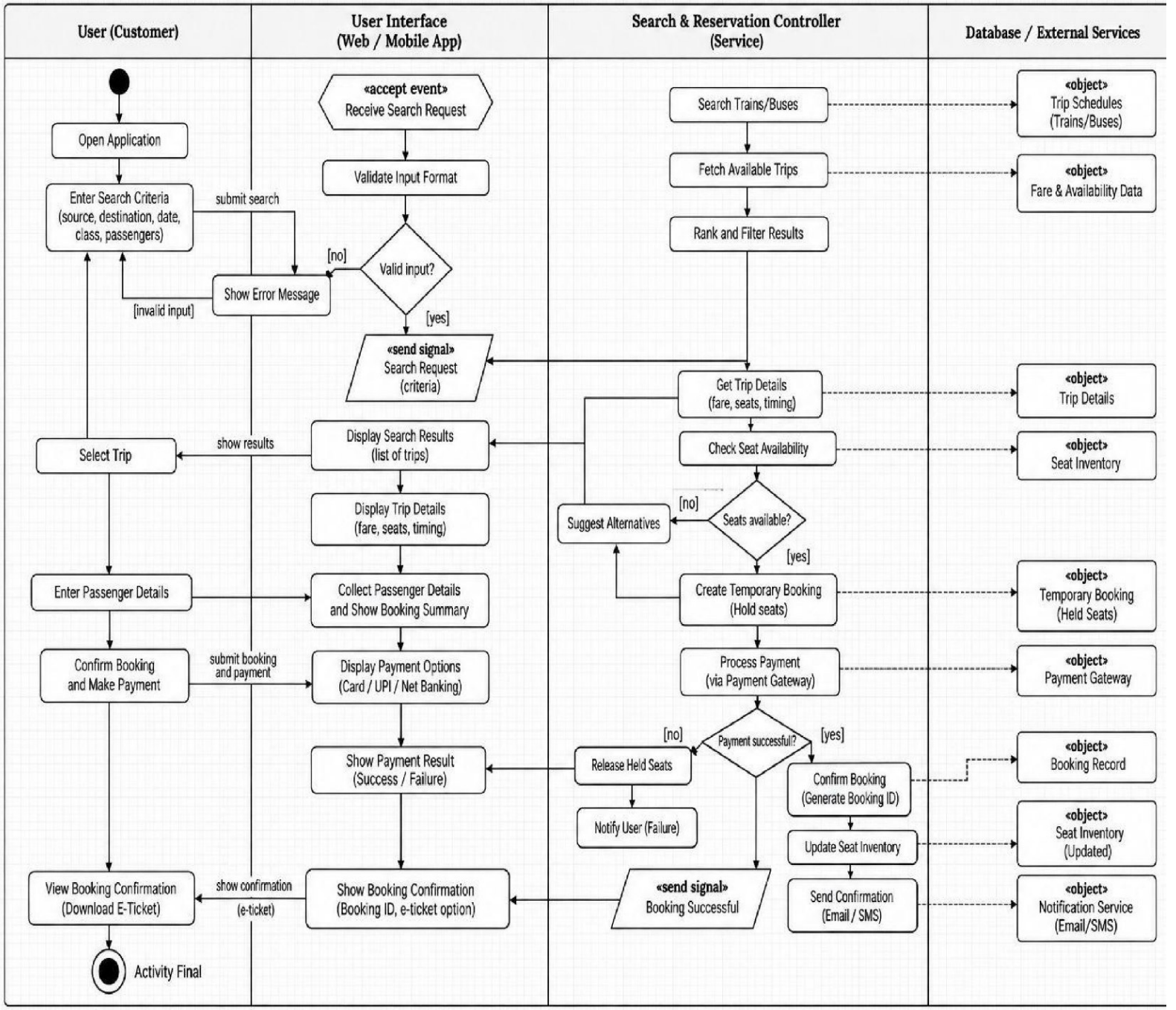
By
25MIS0355 Suhas E
25MIS0349 Nagasanjeevi Balaji N
25MIS0378 Lokesh Kumar G

Activity Diagrams

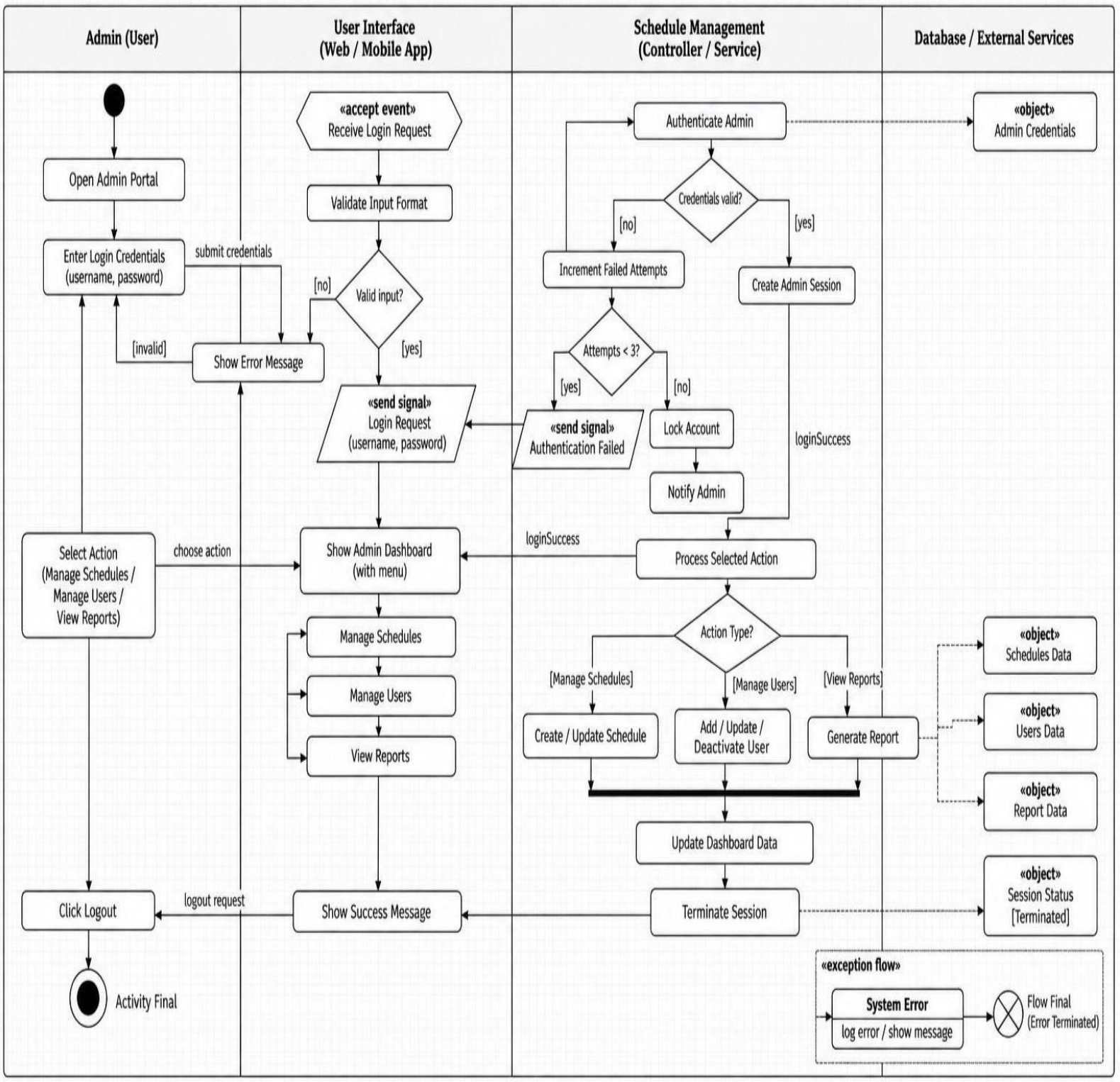
Module 1 – Login & User Management:



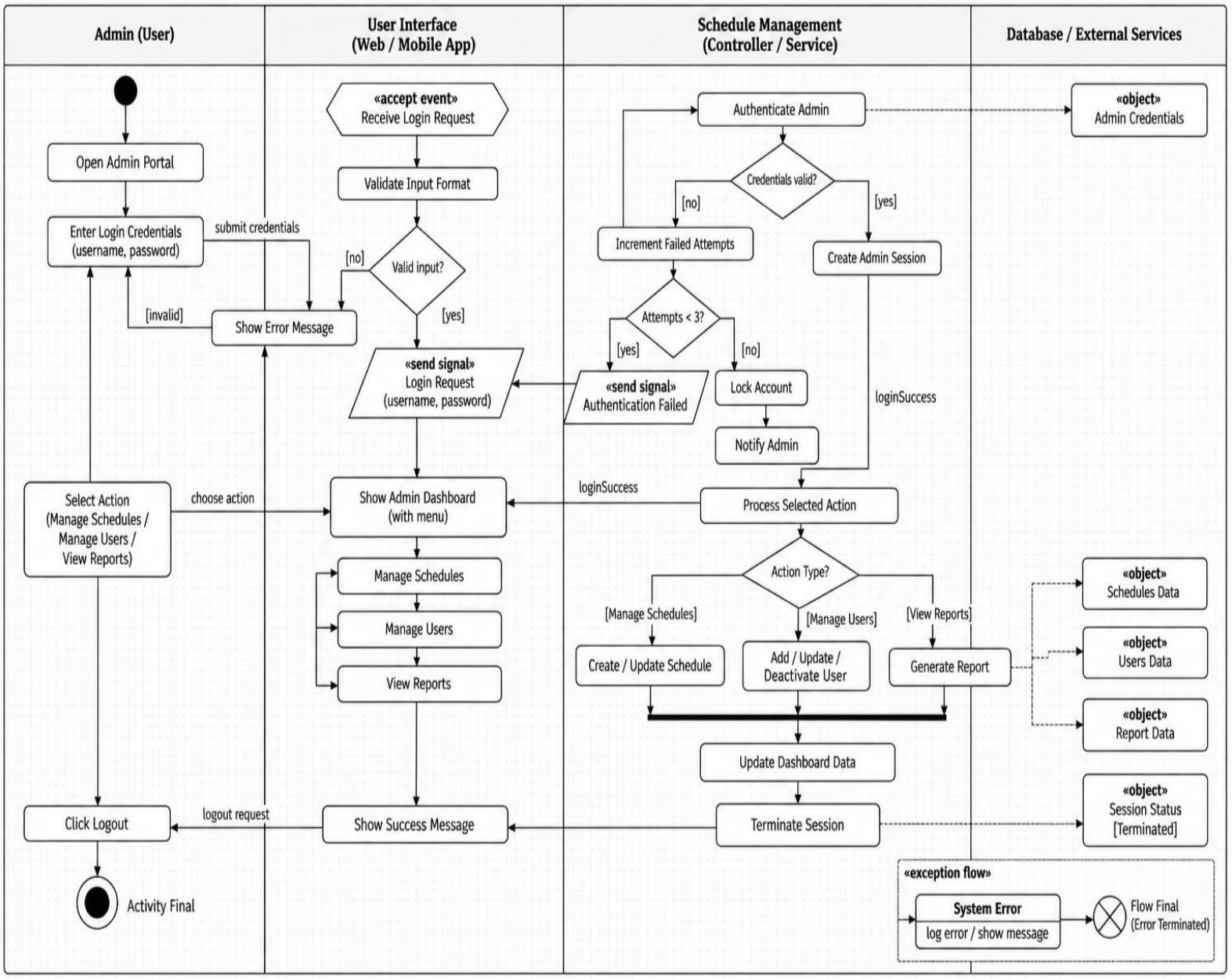
Module 2- Ticket Search & Reservation: 25MIS0378



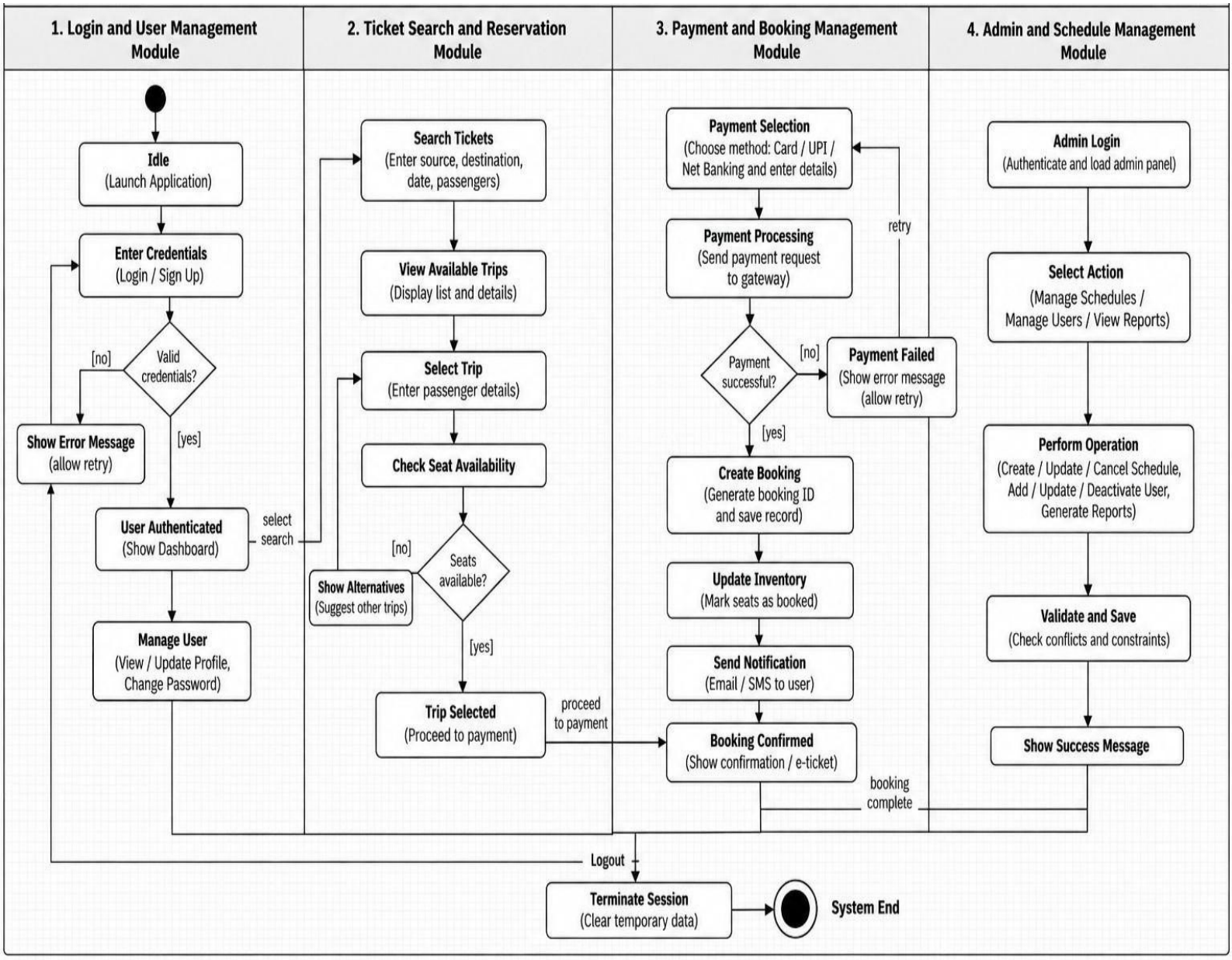
Module 3- Payment & Booking Management: 25MIS0349



Module 4- Admin & Schedule Management: 25MIS0355



Overall Statechart diagram:



State Stimuli Chart:

S.No.	Current State	Stimulus / Event	Next State
1	**Initial State**	System starts	Idle
2	**Idle**	Open application	Enter Credentials
3	**Enter Credentials**	Submit login details	Validate Credentials
4	**Validate Credentials**	Credentials are valid	User Authenticated
5	**Validate Credentials**	Credentials are invalid	Show Error Message
6	**Show Error Message**	Retry login	Enter Credentials
7	**User Authenticated**	Select user management	Manage User
8	**Manage User**	Update profile / change password	Manage User
9	**User Authenticated**	Select ticket search	Search Tickets
10	**User Authenticated**	Logout	Terminate Session
11	**Terminate Session**	Session terminated	Final State
12	**Search Tickets**	Enter source, destination, date and passengers	View Available Trips
13	**View Available Trips**	Select available trip	Select Trip
14	**Select Trip**	Enter passenger details	Check Seat Availability
15	**Check Seat Availability**	Seats are available	Trip Selected
16	**Check Seat Availability**	Seats are unavailable	Show Alternatives
17	**Show Alternatives**	Select alternative trip	Select Trip
18	**Trip Selected**	Proceed to payment	Payment Selection
19	**Payment Selection**	Select payment method and enter details	Payment Processing
20	**Payment Processing**	Payment successful	Create Booking
21	**Payment Processing**	Payment failed	Payment Failed
22	**Payment Failed**	Retry payment	Payment Selection

S.No.	Current State	Stimulus / Event	Next State
23	**Create Booking**	Booking record created	Update Inventory
24	**Update Inventory**	Seats marked as booked	Send Notification
25	**Send Notification**	Confirmation sent	Booking Confirmed
26	**Booking Confirmed**	Ticket generated / booking complete	Module Complete
27	**Module Complete**	Return to dashboard	User Authenticated
28	**Admin Login**	Valid admin credentials	Select Action
29	**Select Action**	Manage schedules	Perform Operation
30	**Select Action**	Manage users	Perform Operation
31	**Select Action**	View reports	Perform Operation
32	**Perform Operation**	Create / update / cancel schedule	Validate and Save
33	**Perform Operation**	Add / update / deactivate user	Validate and Save
34	**Perform Operation**	Generate report	Validate and Save
35	**Validate and Save**	Data valid / operation successful	Show Success Message
36	**Validate and Save**	Invalid data / conflict	Perform Operation
37	**Show Success Message**	Operation completed	Module Complete
38	**Module Complete**	Return to dashboard	Admin Dashboard
39	**User Authenticated / Admin Dashboard**	Logout	Terminate Session
40	**Terminate Session**	Clear temporary data	Final State

State Response Chart:

S.No.	State	State Response / Action
1	**Idle**	Display the application and wait for the user to begin.
2	**Enter Credentials**	Accept username/email and password from the user.
3	**Validate Credentials**	Verify the entered credentials against the registered account.
4	**Show Error Message**	Display an invalid-credential message and allow the user to retry.
5	**User Authenticated**	Create the user session and display the user dashboard.
6	**Manage User**	Allow the user to view/update profile and change password.
7	**Search Tickets**	Accept source, destination, date and passenger details for searching.
8	**View Available Trips**	Retrieve and display available trips with their details.
9	**Select Trip**	Accept the selected trip and passenger details.
10	**Check Seat Availability**	Verify the availability of seats for the selected trip.
11	**Show Alternatives**	Display alternative trips when seats are unavailable.
12	**Trip Selected**	Confirm the selected trip and allow the user to proceed to payment.
13	**Payment Selection**	Display payment methods and accept the selected payment details.
14	**Payment Processing**	Send the payment request to the payment gateway and wait for the response.
15	**Payment Failed**	Display the payment failure message and allow the user to retry.
16	**Create Booking**	Generate a booking ID and save the booking record.
17	**Update Inventory**	Mark the selected seats as booked and update seat inventory.
18	**Send Notification**	Send booking confirmation through email/SMS to the user.
19	**Booking Confirmed**	Display booking confirmation and ticket details.
20	**Admin Login**	Authenticate the administrator and load the admin panel.
21	**Select Action**	Display options for schedule management, user management and reports.

S.No.	State	State Response / Action
22	**Perform Operation**	Create/update/cancel schedules, manage users or generate reports.
23	**Validate and Save**	Check data, conflicts and constraints before saving the operation.
24	**Show Success Message**	Display successful completion of the administrator operation.
25	**Module Complete**	Return the user/admin to the appropriate dashboard after completing an operation.
26	**Terminate Session**	End the active session and clear temporary session data.
27	**Final State**	End the current system interaction.